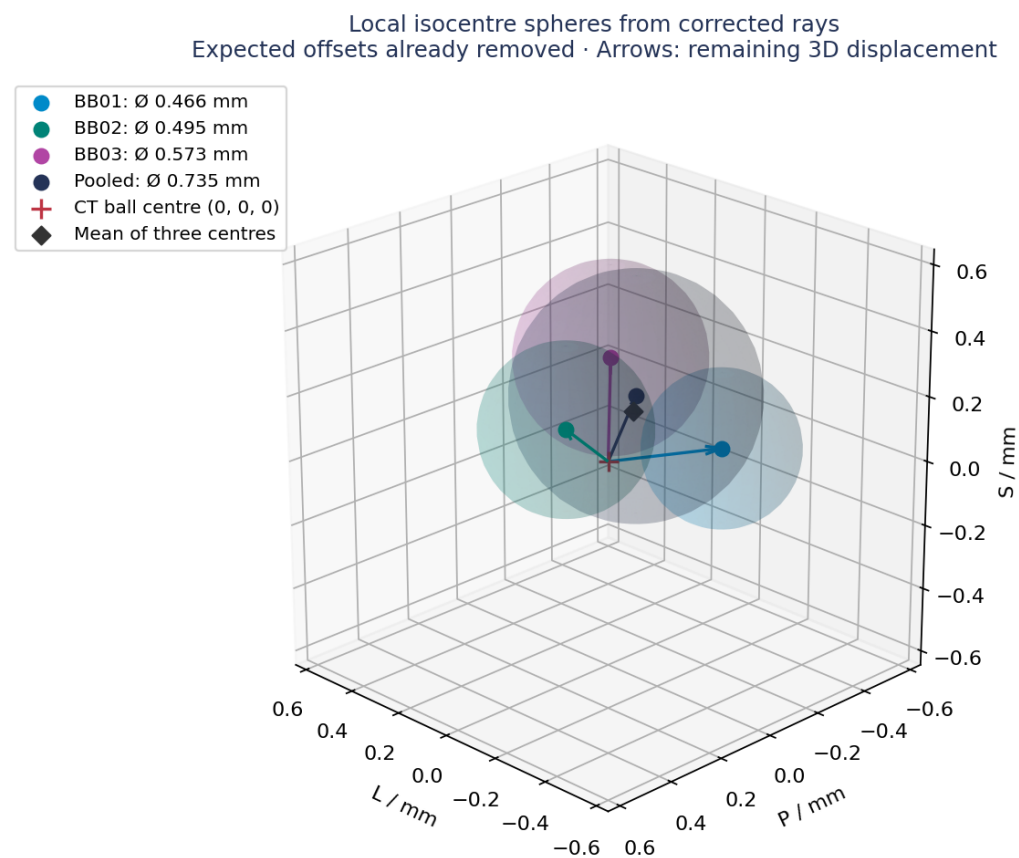


MultiWLT | Results after expected-offset correction



Overall: maximum extra 2D displacement

0.508 mm

Maximum across all balls and images

Pooled envelope diameter

0.735 mm

All corrected rays

12 / 12 ball-field pairs

Acquisition mode: see image export

Analysis status: Review field-centre method

Ball / rays	Envelope diameter / mm	Remaining 3D offset* / mm	Max. extra 2D / mm
BB01	0.466	0.330	0.466
BB02	0.495	0.420	0.508
BB03	0.573	0.343	0.505
All rays pooled	0.735	0.207	0.508

Expected offsets already removed: The planned aperture offset was subtracted as a vector in each image. ***Remaining 3D offset:** Distance from the fitted envelope centre to the red cross (= the corresponding CT ball centre). Arrows show this residual distance. For the pooled fit, each CT ball centre is translated to (0, 0, 0).

Review note: BB03 / G285: Extra displacement 0.231 mm; field-centre method difference 0.697 mm. Two different quantities; explained on page 2.

MultiWLT | Measurements and analysis status

Extra vector = (ball – field)_{MV} – (CT ball projection – planned aperture centre). Subtract X and Y components before calculating the magnitude. 2D values are at the isocentre plane; the 3D envelope uses the local CT ball position.

Gantry	Ball	Expected X / Y (mm)	Observed X / Y (mm)	Extra X / Y (mm)	Extra / mm
20°	BB01	+0.317 / -0.231	+0.718 / -0.468	+0.402 / -0.237	0.466
20°	BB02	+0.115 / -0.082	+0.187 / -0.585	+0.073 / -0.503	0.508
20°	BB03	-0.086 / +0.063	-0.183 / -0.433	-0.097 / -0.496	0.505
105°	BB01	+0.145 / +0.007	+0.141 / +0.105	-0.004 / +0.099	0.099
105°	BB02	-0.116 / -0.006	-0.224 / -0.326	-0.108 / -0.319	0.337
105°	BB03	-0.254 / -0.012	+0.107 / -0.240	+0.362 / -0.228	0.428
195°	BB01	-1.083 / -0.132	-1.219 / +0.046	-0.136 / +0.179	0.224
195°	BB02	+0.024 / +0.003	+0.001 / +0.033	-0.023 / +0.030	0.038
195°	BB03	+0.146 / +0.017	+0.279 / -0.056	+0.134 / -0.073	0.153
285°	BB01	-0.008 / +0.004	-0.090 / -0.081	-0.083 / -0.085	0.119
285°	BB02	-0.480 / +0.549	-0.240 / +0.385	+0.239 / -0.164	0.290
285°	BB03	+0.022 / -0.018	-0.196 / +0.058	-0.218 / +0.077	0.231

BB03 / Gantry 285°: measured extra displacement **0.231 mm**. Separately: **0.697 mm** between the area centroid (search aid only) and opposing 50% edges (field centre used). This flags a method comparison; it is not another ball displacement or an uncertainty estimate.

Supplementary: remaining 3D offset after expected-offset correction. Distance from the fitted local isocentre sphere centre to the CT ball centre of the same target. The expected aperture offset has already been removed; do not subtract it again. This describes the displacement of the envelope, whereas its diameter describes the spread of rays around that centre. Both use the same measured directions; residual positioning errors remain included.

Reference: Expected geometry from the CT and source reference plan, corroborated by delivery records. Return export of the imported RTPLAN remains pending.

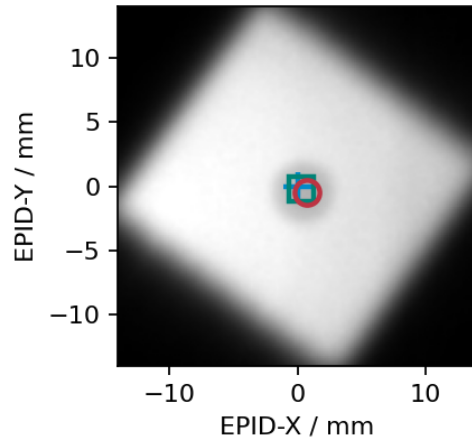
Positioning: confirmed CT reference pose; do not apply setup corrections to the CT ball positions again. No pose fit to the MV images.

Method: pylinac ball detection, field centre from 50% edges; CT ball centres from image data. Results apply to the acquired angles. Review / sign-off: _____

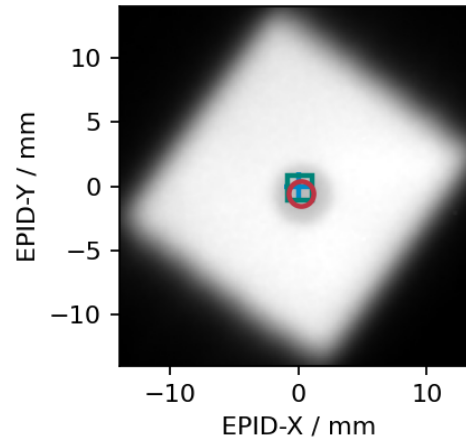
MultiWLT | Gantry 20° · Collimator 54°

Gantry 20° · Collimator 54° | acquired MV image
Blue + Field centre · Green square Expected ball · Red circle Observed ball · Red arrow: extra displacement

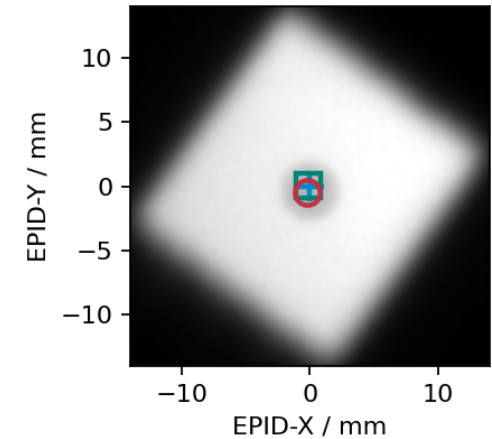
BB01 · additional 0.466 mm



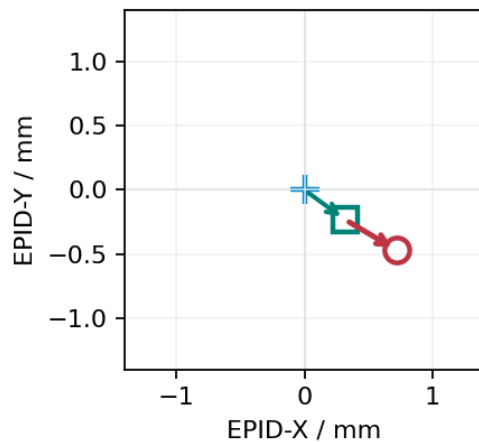
BB02 · additional 0.508 mm



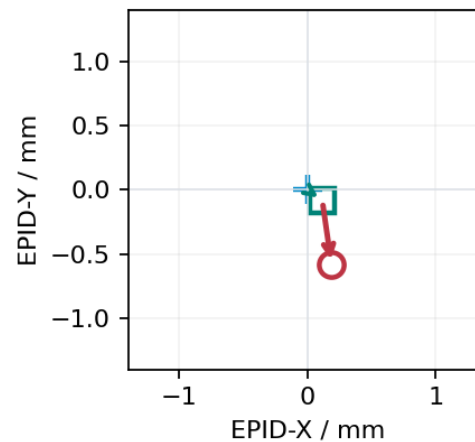
BB03 · additional 0.505 mm



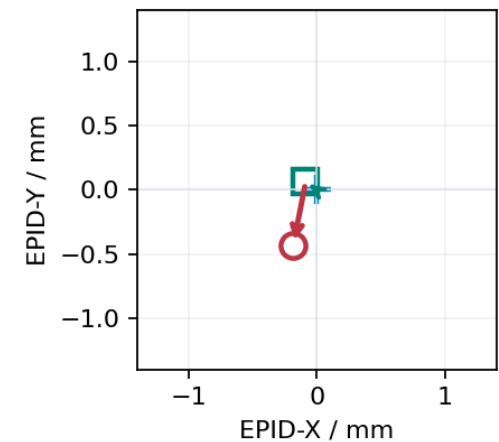
Enlarged centres · Expected 0.392 mm
Observed 0.858 mm · Extra 0.466 mm



Enlarged centres · Expected 0.141 mm
Observed 0.614 mm · Extra 0.508 mm



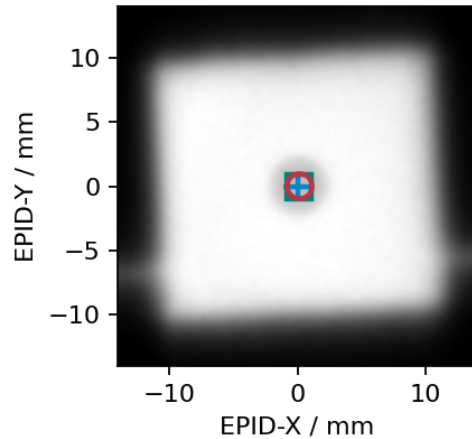
Enlarged centres · Expected 0.107 mm
Observed 0.470 mm · Extra 0.505 mm



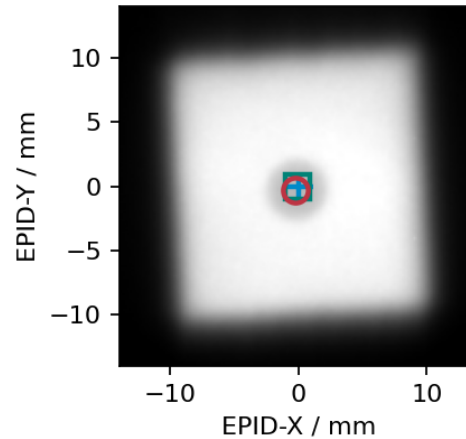
MultiWLT | Gantry 105° · Collimator 93°

Gantry 105° · Collimator 93° | acquired MV image
Blue + Field centre · Green square Expected ball · Red circle Observed ball · Red arrow: extra displacement

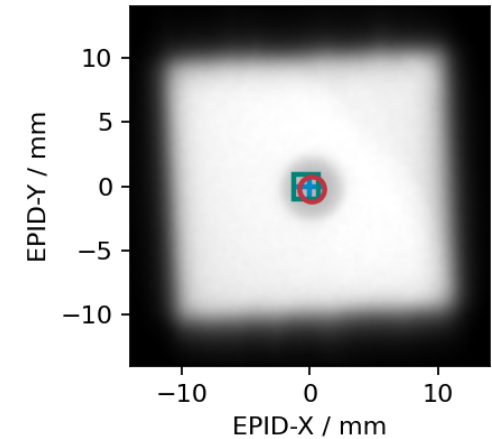
BB01 · additional 0.099 mm



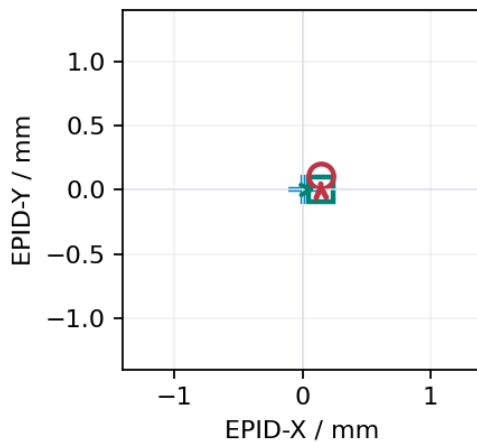
BB02 · additional 0.337 mm



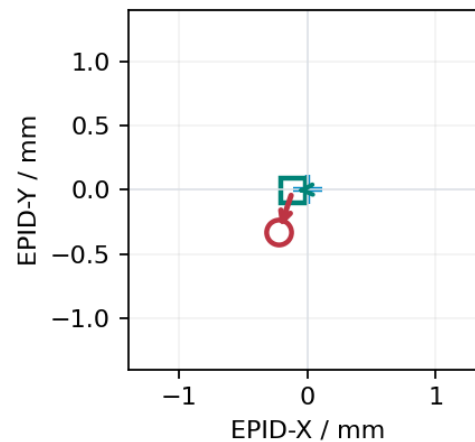
BB03 · additional 0.428 mm



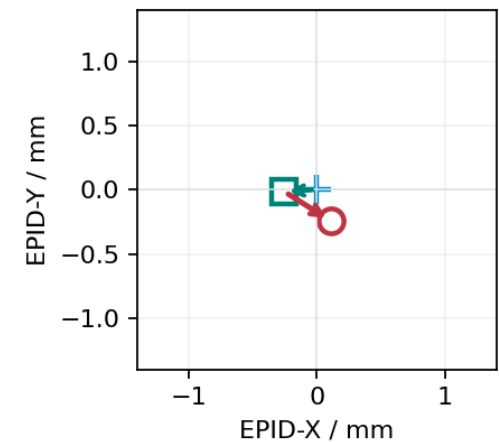
Enlarged centres · Expected 0.145 mm
Observed 0.176 mm · Extra 0.099 mm



Enlarged centres · Expected 0.116 mm
Observed 0.395 mm · Extra 0.337 mm

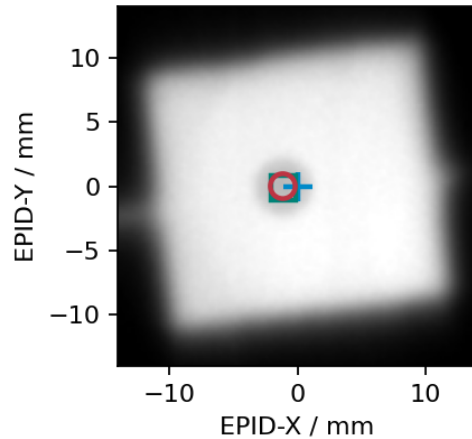


Enlarged centres · Expected 0.255 mm
Observed 0.263 mm · Extra 0.428 mm

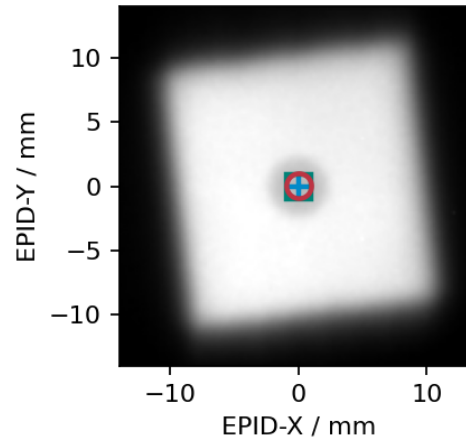


Gantry 195° · Collimator 97° | acquired MV image
Blue + Field centre · Green square Expected ball · Red circle Observed ball · Red arrow: extra displacement

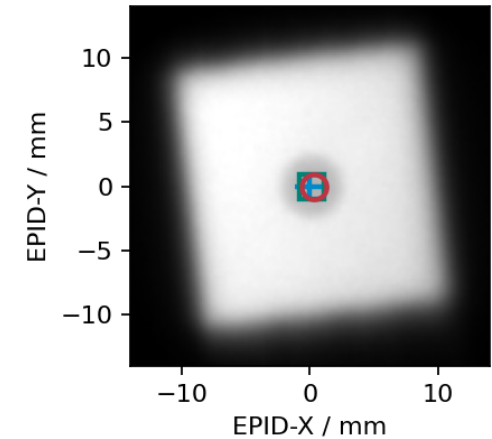
BB01 · additional 0.224 mm



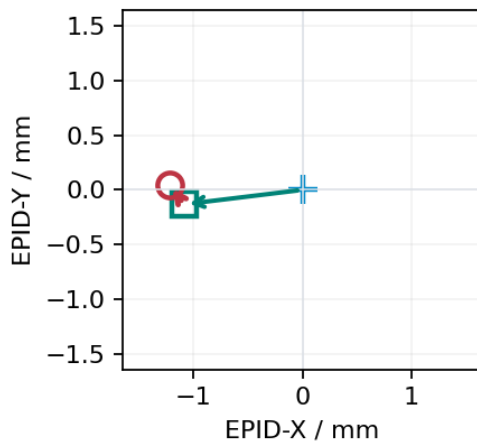
BB02 · additional 0.038 mm



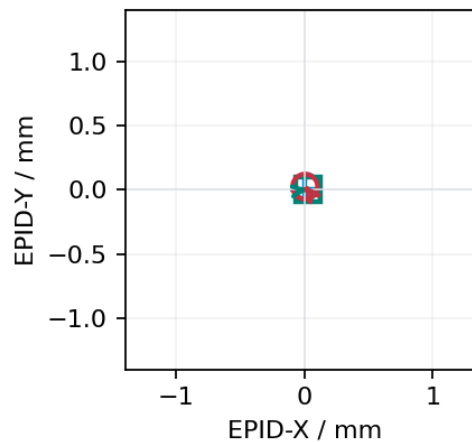
BB03 · additional 0.153 mm



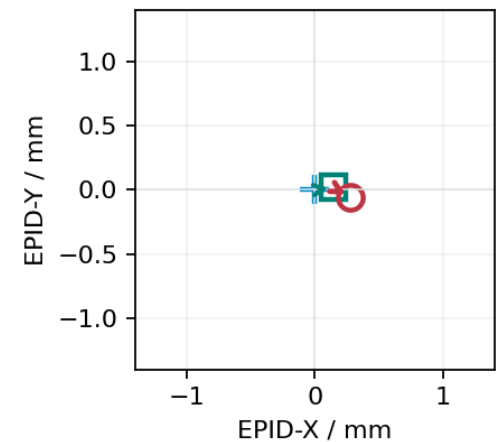
Enlarged centres · Expected 1.091 mm
Observed 1.220 mm · Extra 0.224 mm



Enlarged centres · Expected 0.024 mm
Observed 0.033 mm · Extra 0.038 mm



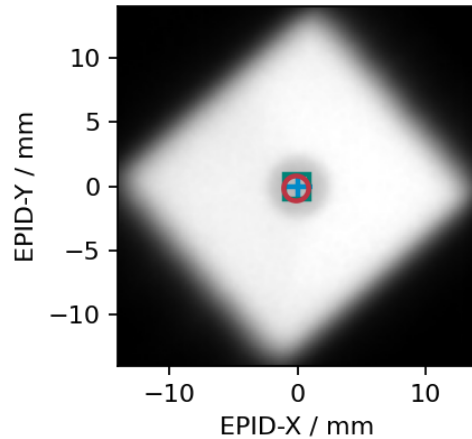
Enlarged centres · Expected 0.147 mm
Observed 0.285 mm · Extra 0.153 mm



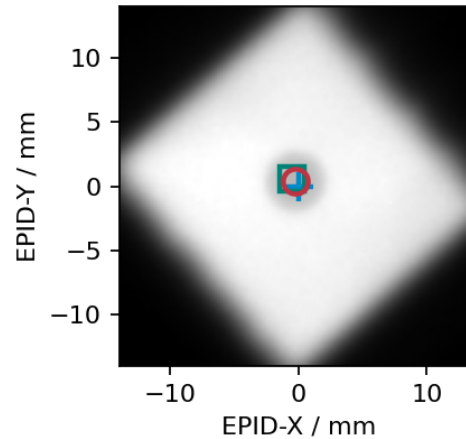
MultiWLT | Gantry 285° · Collimator 41°

Gantry 285° · Collimator 41° | acquired MV image
Blue + Field centre · Green square Expected ball · Red circle Observed ball · Red arrow: extra displacement

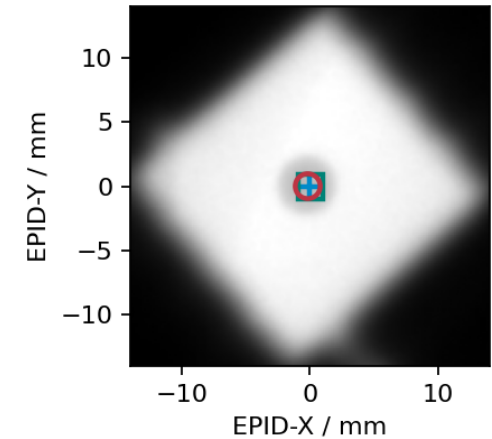
BB01 · additional 0.119 mm



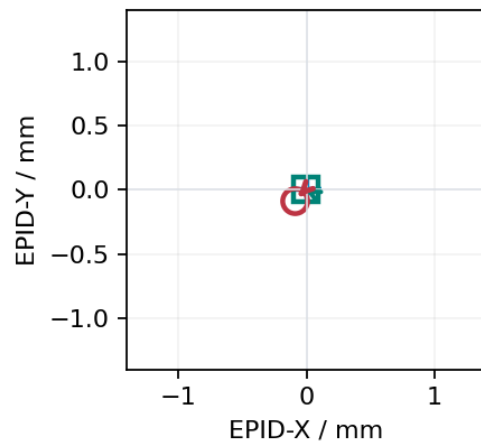
BB02 · additional 0.290 mm



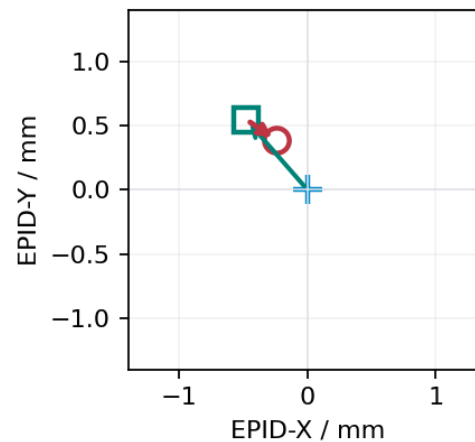
BB03 · additional 0.231 mm



Enlarged centres · Expected 0.009 mm
Observed 0.121 mm · Extra 0.119 mm



Enlarged centres · Expected 0.729 mm
Observed 0.454 mm · Extra 0.290 mm



Enlarged centres · Expected 0.029 mm
Observed 0.204 mm · Extra 0.231 mm

